Glossary

Mineral	a naturally occurring, inorganic solid that has a crystal structure and a definite chemical composition.
Rock	a naturally formed solid made of one or more minerals that make up the earth's crust.
Metallic	consisting of metal.
Moh's Hardness Scale	a scale ranking ten minerals from softest to hardest; used in testing the hardness of minerals.
Luster	the way a mineral reflects light.
Hardness	a measure of how easily a mineral can be scratched.
Streak	the color of a mineral's powder.
Density	the amount of mass in a given volume.
Sediment	small, solid pieces of material that come from rocks.
Weathering	the chemical and physical processes that break down rock at Earth's surface.
Erosion	condition in which the earth's surface is worn away by the action of water and wind.
Deposition	process in which sediment is laid down in new locations.
Fossils	the preserved trace, imprint, or remains of a plant or animal that lived long ago.
Igneous Rock	a type of rock that forms from the cooling of molten rock at or below the surface.
Sedimentary Rock	rock that forms from other rocks or plant/animal remains by compaction and cementation.
Metamorphic Rock	rock that has been changed by heat and pressure.
Rock Cycle	sequence of events in which rocks are formed, destroyed, altered, and reformed by geological processes.
Color	an easily observed physical property used in identifying minerals.

Key Vocabulary

Mineral	A naturally occurring, inorganic solid with a crystalline structure and a definite chemical composition.
Rock	A naturally formed solid made of one or more minerals.
Metallic	Having the appearance or properties of metal.
Mohs Hardness Scale	A relative scale (1–10) used to rank minerals by scratch resistance (talc = 1, diamond = 10).
Luster	The way a mineral reflects light (e.g., metallic, glassy, pearly).
Hardness	A measure of how easily a mineral can be scratched (tested with Mohs scale).
Streak	The color of a mineral's powdered form (produced by rubbing on a streak plate).
Density	Mass per unit volume; helpful for identification (heft test).
Cleavage	Tendency of a mineral to break along flat, planar surfaces.
Fracture	Way a mineral breaks when it does not cleave (uneven, conchoidal, etc.).
Sediment	Small solid pieces of material produced by weathering and erosion.
Weathering	Chemical and physical breakdown of rock at Earth's surface.
Erosion	Movement of weathered material by water, wind, or ice.
Deposition	Laying down of sediment in a new location.
Fossils	Preserved traces or remains of ancient organisms.
Igneous Rock	Rock formed by cooling of molten material (magma or lava).
Sedimentary Rock	Rock formed by compaction/cementation of sediments or of biological remains.
Metamorphic Rock	Rock changed by heat and pressure from a previous rock type.
Rock Cycle	The set of processes that form, change, destroy, and reform rocks.

Key Points and Diagnostic Tests

1. **Mineral vs rock:** Minerals are single, naturally occurring, inorganic, crystalline substances with a fixed or narrowly variable chemical formula. Rocks are aggregates of one or more minerals (or mineraloids). Examples: quartz (mineral) vs sandstone (rock).

2. **Essential identification tests:**

   - *Color:* quick clue but often unreliable alone.
   - *Streak:* color of powdered mineral; more reliable than surface color.
   - *Luster:* metallic, glassy, pearly, etc.; describes light reflection.
   - *Hardness:* use Mohs scale (1–10). Scratch test distinguishes soft vs hard minerals.
   - *Cleavage vs fracture:* look for flat planar breaks (cleavage) versus irregular breaks (fracture).
   - *Density/heft:* relative heaviness can separate similar-looking minerals (e.g., gold vs pyrite).
   - *Other tests:* acid reaction (calcite), magnetism (magnetite), taste/smell only in controlled classroom settings.

3. **Common pitfalls (concept checks from the text):**

- Materials from living organisms (ivory, amber) or man-made materials (Bakelite, Portland cement) are *not* minerals.
- Volcanic glass (obsidian) lacks a crystalline structure and so is usually not classified as a mineral.
- Sandstone is a rock (mixture of minerals like quartz and feldspar), not a single mineral because it is not a pure substance with a fixed formula.

4. **Example identifications from the sheet:**

   - Fluorite: inorganic solid with formula CaF_2; shows glassy luster and white streak $\rightarrow$ qualifies as a mineral.
   - Chrysoberyl: $BeAl_2O_4$, crystalline, hardness about 8.5 $\rightarrow$ mineral.
   - Gold vs pyrite: use hardness and streak. Gold is soft (approx. 2.5–3) and leaves a golden streak; pyrite is harder (6–6.5) and typically leaves a dark/greenish-black streak.

5. **Teaching/assessment notes:** multiple-choice items should emphasize the mineral definition (naturally occurring, inorganic, solid, crystalline, definite composition) and common identification tests. Use short lab tests (streak plate, glass scratch, visual cleavage) to reinforce concepts.